STAT 2020 Discussion Slides

Week 10: Normal Distribution, Linear Combinations, and Sampling Distribution

Shuo Li

University of Iowa

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Index (Outline)

1. Week 10 Goals
2. Standardization and Z-table
3. Percentiles
4. Linear Combinations
5. Linear Expression Probability
6. Sampling Distribution of Xbar

Announcements(0A11 & 0A13)

- Welcome back! We are now **back to normal**.
- The quiz from last week will be released **later this afternoon**.
- We will return **both quizzes together next week**.

Please let me know if you have any questions about grading.

Agenda

- Normal distribution review and Z-table standardization
- Percentiles: find quantiles, and solve for μ or σ
- Mean and variance of linear combinations (assume independence)
- Probability with linear expressions (for example, $P(4X > Y + 1)$)
- Sampling distribution of $\bar{X}$, sums, and related percentiles
- Optional: Delta method preview (only if students ask)
- Attendance check

Standardization Formula

If $X \sim \mathrm{N}(\mu, \sigma^2)$, define

$$Z = \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1).$$

Then

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Z-table usually reports $\Phi(z) = P(Z \leq z)$.

Tail and Interval Forms

$$P(X > x) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right).$$

$$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

Common symmetry:

$$\Phi(-z) = 1 - \Phi(z).$$

Z-table Quick Rules

Most Z-tables report left-tail area:

$$\Phi(z) = P(Z \leq z).$$

So we convert every probability into $\Phi(\cdot)$ form:

$$P(Z > z) = 1 - \Phi(z), \qquad P(a < Z < b) = \Phi(b) - \Phi(a).$$

For negative z, use symmetry:

$$\Phi(-z) = 1 - \Phi(z).$$

Worked Example: Probability Between Two Values

Let $X \sim \mathrm{N}(20, 4^2)$. Find $P(15 < X < 26)$.

$$z_1 = \frac{15 - 20}{4} = -1.25, \qquad z_2 = \frac{26 - 20}{4} = 1.5.$$

$$P(15 < X < 26) = \Phi(1.5) - \Phi(-1.25).$$

Using Z-table:

$$\Phi(1.5) \approx 0.9332, \qquad \Phi(-1.25) \approx 0.1056.$$

$$P(15 < X < 26) \approx 0.8276.$$

Percentile Formula

The p-th percentile x_p solves $P(X \leq x_p) = p$.
For $X \sim \mathrm{N}(\mu, \sigma^2)$:

$$x_p = \mu + z_p \sigma,$$

where z_p is the p-th percentile of $\mathrm{N}(0, 1)$.
Common value:

$$z_{0.975} \approx 1.96.$$

Example: Find the 97.5th Percentile

Suppose $X \sim \mathrm{N}(150, 12^2)$. Find $x_{0.975}$.

$$x_{0.975} = 150 + 1.96 \times 12 = 173.52.$$

So the 97.5th percentile is about

$$173.5.$$

Example: Solve for μ or σ

If $x_{0.975} = 180$ and $\sigma = 12$, then

$$180 = \mu + 1.96(12) \quad \Longrightarrow \quad \mu = 156.48.$$

If μ is known instead, solve

$$\sigma = \frac{x_p - \mu}{z_p}.$$

Inverse Percentile Template (Must Know)

Start from

$$x_p = \mu + z_p\sigma.$$

Then solve by unknown:

$$\mu = x_p - z_p\sigma \quad \text{or} \quad \sigma = \frac{x_p - \mu}{z_p}.$$

Reminder:

$$z_{0.975} \approx 1.96, \qquad z_{0.025} \approx -1.96.$$

Keep the sign of z_p consistent with the percentile p.

Linear Combination Rules

Let $Y = aX_1 + bX_2 + c$.

$$\mathbb{E}(Y) = a\mathbb{E}(X_1) + b\mathbb{E}(X_2) + c.$$

If $X_1 \perp\!\!\!\perp X_2$,

$$\mathbb{V}(Y) = a^2\mathbb{V}(X_1) + b^2\mathbb{V}(X_2).$$

More general form:

$$\mathbb{V}(aX_1 + bX_2) = a^2\mathbb{V}(X_1) + b^2\mathbb{V}(X_2) + 2ab\mathrm{Cov}(X_1, X_2).$$

Example: $Y = 0.8X_1 + 0.2X_2$

Suppose $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{N}(20, 4^2)$, and

$$Y = 0.8X_1 + 0.2X_2.$$

$$\mathbb{E}(Y) = 0.8(20) + 0.2(20) = 20.$$

$$\mathbb{V}(Y) = 0.8^2(16) + 0.2^2(16) = 10.88.$$

Since a linear combination of independent normals is normal,

$$Y \sim \mathrm{N}(20, 10.88).$$

How to Handle $P(4X > Y + 1)$-Type Questions

Rewrite the inequality as one normal variable.
Define

$$W = 4X - Y.$$

Then

$$P(4X > Y + 1) = P(W > 1).$$

If X, Y are independent normals, then W is also normal.

Four-Step Template for Linear Expressions

For expressions like $P(aX + bY > c)$:

1. Define one variable: $W = aX + bY$, so target is $P(W > c)$.
2. Compute moments:

$$\mathbb{E}(W) = a\mathbb{E}(X) + b\mathbb{E}(Y),$$

$$\mathbb{V}(W) = a^2\mathbb{V}(X) + b^2\mathbb{V}(Y) + 2ab\mathrm{Cov}(X, Y).$$

3. If independent, $\mathrm{Cov}(X, Y) = 0$. Then W is normal.
4. Standardize and use Z-table:

$$P(W > c) = 1 - \Phi\left(\frac{c - \mu_W}{\sigma_W}\right).$$

Example: Compute $P(4X > Y + 1)$

Let $X \sim \mathrm{N}(2, 1^2)$, $Y \sim \mathrm{N}(5, 2^2)$, independent.
Set $W = 4X - Y$. Then

$$\mathbb{E}(W) = 4(2) - 5 = 3, \qquad \mathbb{V}(W) = 16(1^2) + 2^2 = 20.$$

So $W \sim \mathrm{N}(3, 20)$, and

$$P(4X > Y + 1) = P(W > 1) = 1 - \Phi\left(\frac{1-3}{\sqrt{20}}\right) = \Phi(0.447) \approx 0.672.$$

Theorem for $\bar{X}$

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2)$, then

$$\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0, 1).$$

Also,

$$S_n = \sum_{i=1}^{n} X_i \sim \mathrm{N}(n\mu, n\sigma^2).$$

Sum-to-Mean Trick

For sums, convert to $\bar{X}$ first:

$$P\left(\sum_{i=1}^{n} X_i > c\right) = P\left(\bar{X} > \frac{c}{n}\right).$$

This often makes standardization cleaner because

$$\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Use this both for probability questions and percentile questions.

Example: Probability for a Sample Mean

Suppose $X_i \overset{\text{iid}}{\sim} \mathrm{N}(20, 4^2)$, $n = 25$. Find $P(\bar{X} > 21)$.

$$\bar{X} \sim \mathrm{N}\left(20, \frac{16}{25}\right), \qquad \mathrm{SE}(\bar{X}) = \frac{4}{5} = 0.8.$$

$$P(\bar{X} > 21) = 1 - \Phi\left(\frac{21 - 20}{0.8}\right) = 1 - \Phi(1.25) \approx 0.1056.$$

Percentiles of $\bar{X}$ and S_n

For $\bar{X} \sim \mathrm{N}(\mu, \sigma^2/n)$:

$$(\bar{X})_p = \mu + z_p \frac{\sigma}{\sqrt{n}}.$$

For $S_n \sim \mathrm{N}(n\mu, n\sigma^2)$:

$$(S_n)_p = n\mu + z_p \sqrt{n}\,\sigma.$$

Same logic: convert to Z, use Z-table, then convert back.

Example: Weight Limit with 0.1% Exceedance

Suppose $X_i \overset{\text{iid}}{\sim} \mathrm{N}(160, 30^2)$, $n = 25$, and

$$Y = \sum_{i=1}^{25} X_i \sim \mathrm{N}(4000, 150^2).$$

Find y_0 such that $P(Y > y_0) = 0.001$.
Equivalent:

$$P(Y \leq y_0) = 0.999 \Rightarrow \frac{y_0 - 4000}{150} = z_{0.999} \approx 3.09.$$

So

$$y_0 \approx 4000 + 3.09(150) = 4463.5.$$

Only If Asked

Delta Method Preview (Optional)

Use when we need approximation for a function of a random variable, $u(X)$.
If X has mean μ_X, variance σ_X^2, and u is smooth near μ_X:

$$\mathbb{E}[u(X)] \approx u(\mu_X) + \frac{u''(\mu_X)}{2}\sigma_X^2,$$

$$\mathbb{V}[u(X)] \approx \left(u'(\mu_X)\right)^2 \sigma_X^2.$$

This is based on Taylor expansion around μ_X.

Practice 1: Standardization

If $X \sim \mathrm{N}(60, 8^2)$, find $P(X > 72)$.

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If $X \sim \mathrm{N}(60, 8^2)$, find $P(X > 72)$.

$$z = \frac{72 - 60}{8} = 1.5, \qquad P(X > 72) = 1 - \Phi(1.5) \approx 0.0668.$$

Practice 2: Percentile

If $X \sim \mathrm{N}(100, 15^2)$, find the 90th percentile.

Practice 2: Percentile

If $X \sim \mathrm{N}(100, 15^2)$, find the 90th percentile. Use $z_{0.90} \approx 1.2816$:

$$x_{0.90} = 100 + 1.2816(15) = 119.22.$$

Practice 3: Linear Combination Moments

Let $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{N}(10, 3^2)$, and $Y = 2X_1 - X_2 + 5$. Find $\mathbb{E}(Y)$ and $\mathbb{V}(Y)$.

Practice 3: Linear Combination Moments

Let $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{N}(10, 3^2)$, and $Y = 2X_1 - X_2 + 5$. Find $\mathbb{E}(Y)$ and $\mathbb{V}(Y)$.

$$\mathbb{E}(Y) = 2(10) - 10 + 5 = 15, \qquad \mathbb{V}(Y) = 2^2(9) + (-1)^2(9) = 45.$$

Practice 4: Linear Expression Probability

If $X \sim \mathrm{N}(1,1)$, $Y \sim \mathrm{N}(2,4)$, independent, compute $P(4X > Y + 1)$.

Practice 4: Linear Expression Probability

If $X \sim \mathrm{N}(1,1)$, $Y \sim \mathrm{N}(2,4)$, independent, compute $P(4X > Y+1)$.

$$W = 4X - Y \sim \mathrm{N}(2,20),$$

$$P(4X > Y+1) = P(W > 1) = 1 - \Phi\left(\frac{1-2}{\sqrt{20}}\right) = \Phi(0.224) \approx 0.589.$$

Practice 5: Sampling Distribution

If $X_i \overset{\text{iid}}{\sim} \mathrm{N}(50, 9^2)$, $n = 36$, find $P(\bar{X} < 52)$.

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If $X_i \overset{\text{iid}}{\sim} \mathrm{N}(50, 9^2)$, $n = 36$, find $P(\bar{X} < 52)$.

$$\mathrm{SE}(\bar{X}) = \frac{9}{\sqrt{36}} = 1.5, \qquad z = \frac{52 - 50}{1.5} = 1.333.$$

$$P(\bar{X} < 52) = \Phi(1.333) \approx 0.9082.$$

Practice 6: Percentile of a Sum

Let $X_i \overset{\text{iid}}{\sim} \mathrm{N}(40, 6^2)$, $n = 16$, and $S = \sum_{i=1}^{16} X_i$. Find $s_{0.95}$, the 95th percentile of S.

Practice 6: Percentile of a Sum

Let $X_i \overset{\text{iid}}{\sim} \mathrm{N}(40, 6^2)$, $n = 16$, and $S = \sum_{i=1}^{16} X_i$. Find $s_{0.95}$, the 95th percentile of S.

$$S \sim \mathrm{N}(16 \times 40,\ 16 \times 6^2) = \mathrm{N}(640, 24^2).$$

Using $z_{0.95} \approx 1.645$:

$$s_{0.95} = 640 + 1.645(24) = 679.48.$$

Class Reminder: Exam 2 Questions

- We can review concepts and methods from homework and quizzes.
- Please do not discuss any actual Exam 2 question in class.
- Students can discuss exam-specific questions during office hours.

Take-Home Messages

- Standardize first; then use the Z-table.
- Percentiles follow $x_p = \mu + z_p\sigma$ (or its inverse forms).
- For independent variables, linear-combination variance adds by squared coefficients.
- Convert expressions and sample means into one normal variable before computing probability.

In-Class Attendance Check (Quick)

Suppose $X_1, \ldots, X_{25} \overset{\text{iid}}{\sim} \mathrm{N}(70, 10^2)$.
What is

$$P(\bar{X} < 73)?$$

Write your answer in $\Phi(\cdot)$ form only (no decimal approximation).

Please submit: Full Name; Section Number; Your Answer

In-Class Attendance Check (Quick)

Suppose $X_1, \ldots, X_{25} \overset{\text{iid}}{\sim} \mathrm{N}(70, 10^2)$.
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Write your answer in $\Phi(\cdot)$ form only (no decimal approximation).

Please submit: Full Name; Section Number; Your Answer

Answer:

$$\bar{X} \sim \mathrm{N}\left(70, \frac{100}{25}\right), \quad z = \frac{73 - 70}{10/5} = 1.5,$$

$$P(\bar{X} < 73) = \Phi(1.5).$$